is off by 2 bps, the value of the error is around $2 \times \$182{,}500 = \$365{,}000$.[5] □

It should be noted that PV01 is not a perfect concept; the swap curve typically does not move in a parallel fashion (this is one of the motivating factors why some investors prefer to decouple the fixed leg from the floating leg of a swap and look at the PV01 without LIBOR set). Some investors will look at their exposure to a shift in individual swap rates for more granular information on their swap or portfolio of swaps.

2.2 Spread Risk

In the previous section, we considered the risk inherent in a swap given a shift in swap rates. Note that when a swap rate moves 1 bp, this can happen in any number of ways. For example, if the 5-year swap rate were to go up 1 bp, this could occur if the 5-year Treasury yield increased 1 bp, or if the 5-year swap spread increased 1 bp, or if the 5-year Treasury yield decreased 1 bp and at the same time the 5-year swap spread increased 2 bps, etc. Despite the fact that a movement in swap spreads is subsumed in a movement in swap rates, it is customary to consider the sensitivity that swaps have to the movement of swap spreads in isolation.

Consider now the market as depicted in Table 1.1 and a \$100mm 5-year swap in which a client received at a rate of $1.606\% = T_5 + 17$. That is, at the time this swap was executed, the bid side swap spread was 17 bps, and the 5-year note was yielding 1.436%. Suppose that a moment after the swap is executed, 5-year Treasury yields remain the same, but 5-year swap spreads fall 1 bp to 16. In this case, the client has received fixed at 1.606%, and the 5-year swap rate is now only 1.596%. The swap is now in-the-money to the client. The client will have recognized a mark-to-market gain on the swap. How much is this worth to the client? According to the PV01 of a 5-year swap found in Table 2.3, the client will be up \$40,237.[6]

Alternatively, suppose that after executing at 1.606%, 5-year Treasury yields are unchanged, but 5-year swap spreads increase 1 bp to 18. Now the client finds that he has received at 1.606% and 5-year swap rates are 1.616%. In this case, the swap is out-of-the-money to the client. And once again, the

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[5] In Section 1.2.1, we said the value of this mistake was around \$365,000, and now we see why.

[6] This is not exactly what the PV01 of a swap tells us. PV01 considers a parallel shift in the curve. Here we have considered only a movement in the 5-year swap spread. The two numbers are practically the same, however. That is, the contribution to the PV01 of an n-year swap comes almost entirely from the movement in the n-year swap rate and very little from the movement in other rates.

PV01 of the swap will tell us that the client is down $40,237.[7] As these examples indicate, swaps have spread risk, meaning that their value is affected even by movements in swap spreads alone. That is, even when the relevant Treasury yield is unchanged, the value of a swap changes when swap spreads alone move.

We can actually take this realization a step further. If we pay fixed in a swap and buy Treasuries to hedge, the net position is such that we are not exposed to movements in Treasuries but are only exposed to movements in swap spreads. Let's go back and take a look again at Figure 1.4, which shows the position of a dealer who pays fixed and hedges himself by buying Treasuries. The dealer has paid fixed at $T_5 + 17 = 1.436\% + 17$ bps $= 1.606\%$ and has purchased Treasuries at a price of 101-16. Equivalently, we can say that the dealer purchased Treasuries at a yield of 1.436%. Let's look at the value of the swap plus the hedge both in the case in which the 5-year Treasury yield, T_5, moves and in the case in which the 5-year swap spread, denoted ss_5, moves.

As shown in Table 2.4, a change in the Treasury yield by 1 bp changes the value of the swap and the value of the Treasury hedge in equal but opposite directions. Thus, the value of the net position is unchanged given a change in Treasury yields. This will be true for any movement in Treasury yields. However, a move in swap spreads changes the value of the net position: when spreads go up, the trade moves in favor of the dealer, and when spreads go down, the trade moves against the dealer.

This type of position in which one enters into a swap and trades Treasuries in order to hedge the interest rate risk associated with a fixed-floating swap is referred to as a *spread trade*.[8] If someone receives fixed and sells Treasuries to hedge, this is referred to as being *short spreads* or having *sold spreads*. In this trade, the value of the position will go up if spreads go down and go down if spreads go up. Movements in Treasury yields do not affect the value of the position.

If one pays fixed and buys Treasuries to hedge, this is known as being *long spreads* or having *bought spreads*. In this case, an increase in spreads

[7] As mentioned previously, PV01s are not constant but depend upon the level of rates. Thus, for even a slight move of a single basis point in the swap rate, the PV01 will not be the same in both directions. However, the magnitude of the difference is so small that it can be safely ignored for our purposes here.

[8] There are many common trades in other areas of the fixed income markets in which the interest rate risk inherent in a position is hedged with Treasuries. For example, consider an investor who purchases a fixed rate corporate bond. The investor is exposed to both changes in interest rates and changes in the underlying credit. Oftentimes investors just want to take a view on the underlying credit but not on interest rates. In this case the investor who bought the fixed rate corporate bond will sell Treasuries of a comparable maturity. The investor is immunized against changes in the level of rates and is only exposed to changes in the underlying corporate spread.

	Pay Fixed at 1.606% and Buy Treasuries at a Yield of 1.436%		
Variable	Change in Swap Value (bps Running)	Change in Treasury Value (bps Running)	Net (bps Running)
$T_5 \uparrow 1$ bp	+1bp	−1bp	0bp
$T_5 \downarrow 1$ bp	−1bp	+1bp	0bp
$ss_5 \uparrow 1$ bp	+1bp	0bp	+1bp
$ss_5 \downarrow 1$ bp	−1bp	0bp	−1bp

Table 2.4: Change in Value of Spread Trade Given Movement in Underlying Treasury and Swap Spread

means that the value of the position will go up, and a decrease in spreads will mean a loss. A movement in the Treasury yields does not affect the value of the position. Swap spreads are a very popular thing to trade in the swaps market. People put on positions expressing a view on (or hedging against) movements in swap spreads. Those who wish to express the view that spreads will widen will buy spreads, and those who wish to express the view that spreads will tighten will short spreads.

2.2.1 A Closer Look at Swap Spreads

Figure 2.2 shows historical mid-market (defined to be the average of the bid and offer) swap spreads for 2-year and 10-year swaps. Once again, the 2-year swap spread is the spread in basis points added to the yield of the on-the-run 2-year Treasury to come up with the 2-year swap rate. The 10-year swap spread is defined analogously. Notice that 2-year spreads are typically less than 10-year spreads.[9] Swap spreads were fairly stable in the early to mid-1990s. As we approached the end of the millennium, and Treasury supply was diminishing and concerns about Y2K (year 2000) were rampant, swaps spreads were generally increasing. Since the year 2000 up until the financial crisis, swaps spreads have generally trended lower. The market saw a dramatic widening of 2-year spreads during the crisis, but since that time spreads have generally tightened. Following the phenomenon of 30-year

[9] In late 1994, 2-year swap spreads traded at levels above 10-year spreads for several days when it had been announced that Orange County's investment portfolio had gone bankrupt. Orange County had invested heavily in structured notes, bonds whose coupons are linked to underlying derivative transactions. As we will see in Chapter 7, the market correctly responded to this news with an immediate increase in 2-year, but not 10-year, swap spreads.

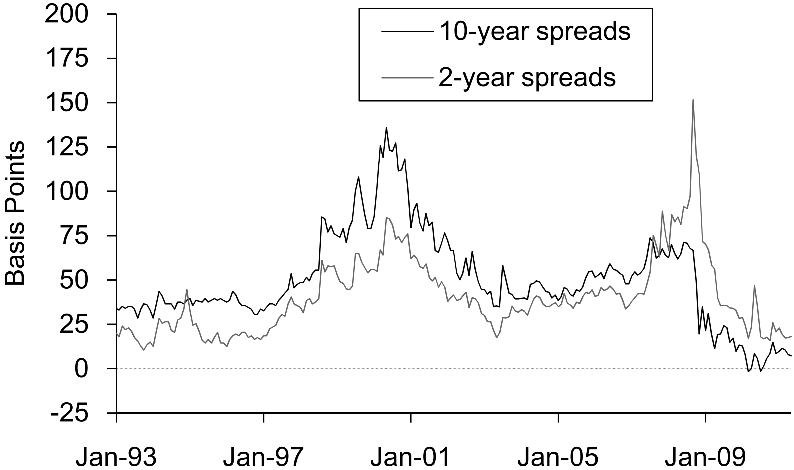

Figure 2.2: Historical Swap Spreads

spreads trading in negative territory (see Chapter 1), 10-year spreads first traded in negative territory in the spring of 2010.

There are many competing factors that explain either the equilibrium level of swap spreads or movements in swap spreads from one day to the next. Invariably, one of these factors takes center stage at any given time, receiving the attention of market participants as the leading explanation of the primary driver of spreads. Let's first start with an explanation suggesting the equilibrium level of swap spreads:

- **The GC-LIBOR Argument.** Let's go back to Figure 1.4, which shows what happens when a dealer pays fixed and hedges himself by buying Treasuries and enters into a repo transaction to finance the Treasury position. But let's redraw this figure, focusing only on the rates that are either paid or received over the life of the swap, as well as the yield on Treasuries that is earned throughout time, and the repo rate that is paid out over time as the dealer finances the Treasury purchase. In other words, let's redraw Figure 1.4, excluding the initial purchase of Treasuries and payment for those Treasuries, and just concentrate on the flows that occur over the life of the trade. This is demonstrated in Figure 2.3. By buying 5-year Treasuries, the dealer is earning the yield T_5. He needs to finance this purchase through a repo and pays a rate of interest equal to the GC (general collateral)

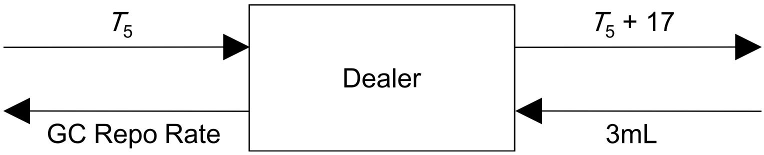

Figure 2.3: The GC-LIBOR Explanation of Swap Spreads

repo rate.[10] The repo typically lasts for a relatively short period of time, either overnight or for *term* (i.e., more than one day), but the point here is that the repo rate is not locked in over the full five years of the trade and is likely to be renegotiated from time to time, depending on then-prevailing repo rates.[11] In the swap itself, the dealer of course pays a fixed rate equal to $T_5 + 17$ and receives LIBOR, which will change every three months.

So let's put this all together: from the top two arrows, we see that the dealer is earning T_5 on the Treasuries and paying out $T_5 + 17$, for a net position of paying out the swap spread of 17 bps. From the bottom two arrows, we see the dealer is earning a net spread equal to the difference between LIBOR and GC repo rates, which is typically a positive number.[12]

A dealer who enters into this trade should on average expect to break even, and thus the GC-LIBOR explanation of swap spreads suggests that the swap spread should be equal to the expected differential between LIBOR and GC rates over time. Of course, LIBOR is likely to change over time, and GC repo rates are likely to change, and their difference is likely to change as well. If we were absolutely certain that the difference between LIBOR and GC rates were to remain constant over time, then this differential should be equal to the swap spread.

Now let's look at factors that don't necessarily suggest the equilibrium level of swap spreads but definitely cause swap spreads to move from one

[10]Or possibly a lower rate if the Treasuries are trading special. We will ignore that possibility in this discussion.

[11]Moreover, when hedging this swap, many dealers would choose not to maintain a long position in this particular Treasury the entire five years but would rather roll into the on-the-run whenever a new issue settles in order to maintain a position in the most liquid issue. We will ignore this possibility as well.

[12]Recall that repo rates are generally lower than LIBOR rates of comparable maturities: repos are secured loans, whereas LIBOR represents the rate of interest banks charge other banks for unsecured loans.

day to the next, or even one minute to the next minute. It is important to note that these factors may not occur in isolation, and ferreting out which factor is the dominating effect as it pertains to movements in spreads is often difficult.

- **Supply and Demand.** This is the biggest factor that determines swap spread movement from one moment to the next. If investors in aggregate want to pay fixed to dealers, this will increase swap spreads, and as they want to receive fixed in aggregate, this will tend to decrease swap spreads.[13]
- **Treasury Supply.** Depending on the funding needs of the U.S. government, the U.S. Treasury may either increase or decrease future issuance of Treasuries. Even the expectation of a shift in the supply of Treasuries can have a dramatic effect on the market. If investors believe for some reason that the supply of newly issued Treasuries is going to be lower than had been previously expected, this is going to raise the demand for existing Treasuries, raising their prices, and lowering their yields. This is an example of a *technical bid* in the market for Treasuries: Treasuries prices may increase and yields may decrease not due to some fundamental outlook on the economy but rather on supply and demand considerations. As Treasuries become a scarce resource, the demand for them will increase. Although Treasury yields will naturally decline in this scenario, this should not necessarily have any bearing on swap rates. Swap rates are a function of LIBOR rates. Inasmuch as a decrease in Treasury yields due to the diminishing supply of Treasuries may not affect (current or future anticipated) LIBOR rates, this would suggest that swap spreads may widen as Treasury supply decreases.
- **The Shape of the Yield Curve.** The steeper the yield curve (i.e., the greater the differential between long-term rates and short-term rates), the more incentive investors will have to receive fixed in, say, a 10-year swap (a long-term rate) and pay 3-month LIBOR (a short-term rate). As we will see later in this chapter, this dynamic is a key driver inducing corporate issuers to receive fixed in swaps in order to swap fixed rate issuance into a synthetic floating rate issuance. As discussed previously, if investors on aggregate want to receive fixed in swaps, this

[13]The dynamic described here is similar to, say, the market for a stock. As investors in general want to buy a stock, this will tend to push the price up. Similarly, as investors want to pay fixed, this will tend to push up swap spreads, as dealers will want to receive a higher fixed rate (which is composed of the Treasury yield plus the swap spread) for additional swaps that they trade.

will tend to cause swap spreads to tighten. Thus, as the yield curve steepens, swap spreads often tend to tighten.[14]

- **The Level of Rates.** The conventional wisdom is that swap spreads should tighten as the bond market rallies (provided the rally is not a flight to quality) and widen as the bond market sells off. As the bond market rallies, mortgage homeowners have an increased incentive to refinance their mortgages. Holders of mortgage securities therefore tend to experience prepayment of their mortgages as the market rallies. In order to hedge this risk, those who hedge their mortgage holdings will tend to receive fixed as the market rallies. Conversely, as the market sells off and mortgage securities prepay slower than expected and therefore extend, holders of mortgage securities will often pay fixed in swaps to hedge.[15]

- **Systematic Credit and Liquidity Concerns.** A shock to the financial system of almost any sort (e.g., Russia defaulting on its sovereign debt, the near collapse of Long-Term Capital, terror threats, Y2K concerns, etc.) is typically accompanied by a "flight to quality" in which investors shed assets with credit risk and buy risk-free Treasuries. LIBOR rates, which represent the rates that banks lend to one another, embody some component of credit risk and as such tend to rise as perceived systemic risk to the financial system is heightened. Thus, in such an event, Treasury yields decline, LIBOR rates rise, and this tends to increase swap spreads.

- **Repo Rates.** Look back at Figure 1.4, which shows the position of a swap dealer paying fixed, buying Treasuries to hedge, and entering into a repo to finance the Treasuries. Suppose that ceteris paribus repo rates decline. This could occur if, say, the on-the-run five-year Treasury were to go special. In this case, the dealer is saving on his interest expense in the repo and would be able to afford (i.e., would still break even) if he were to pay a higher swap spread. Thus, as repo rates go down, swap spreads tend to go up, and vice versa.

  The magnitude of this effect tends to be quite small and dissipates as the maturity of the swap increases. This is due to the fact that a change in an overnight repo rate is an effect that has an impact for only one day, whereas any change in a swap spread impacts over the life of the swap.

[14] This is not to suggest that swap spreads always tighten when the curve steepens. Other market forces may be at work that cause spreads to widen when the curve is steepening.

[15] For a quick review on mortgages, see Appendix D.

Example

Consider the example depicted in Figure 1.4 and assume a swap dealer paid fixed on \$100mm 5-year swaps at a rate of 1.606% and hedged the position by buying \$100mm 5-year notes at 100-16. As shown in Equation 1.13, the dirty price of the bond is \$101,557,065.22, and based upon Equation 1.14, under the assumption that the overnight repo rate was 20 bps, the swap dealer would owe \$101,557,629.43 upon financing the position for one day. If instead the repo rate were 0 bps, the swap dealer would instead owe \$101,557,065.22 after financing the position for a day (i.e., the amount owed is equal to the amount borrowed when the repo rate is zero). Thus, a 20 bps change in the repo rate reduces the cost of financing by \$101,557,629.43 − \$101,557,065.22 = \$564.21. Moreover, assume that it is anticipated that the repo rate on this security will fall from 20 bps to 0 bps until August 31, 2010 (a total of 19 days), the settlement date of the next newly issued 5-year note. In this case, the financing costs of the dealer will be expected to be reduced by \$564.21 × 19 = \$10,719.99. Suppose this savings to the swap dealer were to be reflected in an increased swap spread paid by the dealer.[16] The PV Coupon 01 in this swap can be approximated by putting

$$\begin{aligned} \text{PMT} &= 1 \text{ bp}/2 \\ n &= 10 \\ i &= 1.606\%/2 \end{aligned}$$

and getting PV = 4.8 bps, or around \$48,000 on the notional of \$100mm. Thus, the reduced financing costs of \$10,719.99 translated into a 5-year swap spread increase of

$$\frac{\$10{,}719.99}{\$48{,}000 \text{ per bp}} \approx 0.2 \text{ bps.} \qquad \square$$

The Summer of 2003

The summer of 2003 was a particularly volatile time in the fixed income markets and was a period in which supply and demand factors greatly influenced the level of Treasury yields and swap spreads. Figure 2.4 shows

[16] The forces of competition will incentivize a swap dealer to increase the swap spread he is willing to pay when repo rates decline. If repo rates in the market drop, then all swap dealers who are paying fixed and buying Treasuries to hedge would benefit from reduced financing costs. In order to win more business in this scenario, swap dealers would be inclined to increase the spread they are willing to pay, without affecting their P&L (profit and loss).

movements in 10-year swap spreads superimposed on movements in 10-year Treasury yields during this period. Over the course of six weeks, the yield of the on-the-run 10-year Treasury increased nearly 150 bps.[17] As yields on Treasuries began to rise, the duration of mortgage securities increased as well.[18] This meant that these securities were becoming even more sensitive to changes in interest rates, causing large mortgage holders to rebalance their portfolios.[19] Some estimate that over this six-week period, the aggregate duration of the mortgage market increased by $800bn 10-year equivalents (i.e., loosely speaking, it was as if mortgage investors in aggregate had gotten long an *extra* $800bn 10-year Treasuries during a six-week period in which the market had sold off about 150 bps).[20] Investors who had huge portfolios of mortgages, needing to manage their interest rate risk, found it useful to do so either by selling Treasuries or by paying fixed in swaps.[21] As more and more mortgage holders rebalanced their portfolios by selling Treasuries or by paying fixed in swaps, this actually fed on itself, causing rates to rise further, which triggered additional selling.[22] As swap dealers were asked to receive fixed more and more, swaps spreads started to rise—and did so rapidly. Both the vast selling of Treasuries and the paying of fixed in swaps caused Treasury yields and swap spreads to rise dramatically. After a number of weeks, the selling died down, as did the need for mortgage investors to pay

[17]There were a number of reasons for the dramatic sell-off in the market, but to many it is believed that the initial catalyst was the announcement by the Fed that they were *not* going to purchase Treasury securities in the open market as an unconventional method of maintaining low interest rates in order to stimulate the economy. There had been much speculation that the Fed would resort to this unconventional method, and investors who had held Treasuries in part anticipating that the Fed would be forced to buy Treasuries in this manner were quick to sell them upon the announcement by the Fed, thus causing Treasury yields to begin to rise rapidly.

[18]The value of mortgage securities is sensitive to the level of interest rates. Mortgage securities are subject to prepayment, meaning that the underlying loans can be prepaid by the borrower. As interest rates fall, the economic incentive for borrowers to refinance increases, which is to say that their incentive to prepay their existing mortgages increases. As rates rise, it becomes less likely that the loans will be prepaid, and hence the average life of a mortgage, and its duration, increases as well. Once again, for a review of mortgages, see Appendix D.

[19]This is the characteristic negative convexity of mortgages: as rates rise, mortgages increase in duration. Thus, if we hold mortgage securities in our portfolio, our portfolio becomes "longer the market" just when we don't want that to occur.

[20]The movements in Treasury yields during this time were not uniformly in one direction; the market experienced significant oscillation over this period, sometimes rallying and sometimes selling off before finally reaching equilibrium levels at much higher yields some weeks later.

[21]Remember, we said that paying fixed in a swap is kind of like selling a fixed rate bond.

[22]The adage "selling begets selling" definitely applies.

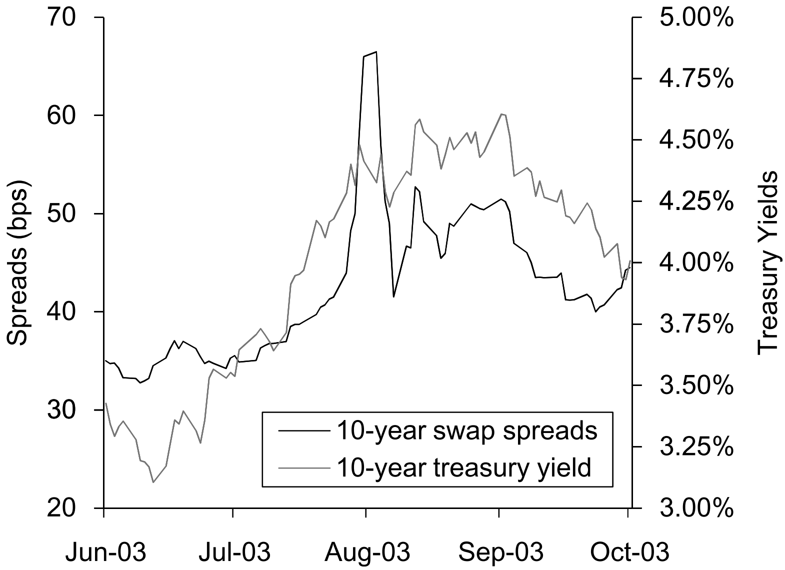

Figure 2.4: Swap Spreads Versus Treasury Yields During the Summer of 2003

fixed.[23] As the smoke cleared, the swaps market had found that 10-year swap spreads had been through one of their most dramatic increases anyone had ever seen. At this point, speculators came into the market and put on positions that expressed their view that swap spreads would soon decrease back within historical norms. As Figure 2.4 shows, spreads did return to lower levels shortly thereafter.

2.3 Currency Risk

Cross-currency swaps have currency (or exchange rate) risk. The easiest way to see this is to recall that the initial and final exchanges of principal are made at the spot exchange rate. If exchange rates move dramatically over the life of the transaction, the off-marketness of the final payments at maturity can be huge.

[23]The selling died down, as did the need for mortgage investors to pay fixed once the rate of mortgage extension slowed significantly. That is, after the market had sold off to the extent that it had, any further increase in rates would not cause mortgage durations to increase substantively. To put this another way, given the large backup in rates, the embedded options to prepay found in mortgage-backed securities (MBS) were so far out-of-the-money that the negative convexity in such securities was of little consequence.